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CS5400: Introduction to Artificial Intelligence

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PROJECT

Inferring and Regulating Cognitive States Using EEG and Artificial Intelligence

Motivation: Understanding and regulating human cognitive states (e.g., focus, stress, relaxation) is a central challenge in AI-driven human–machine interaction. Wearable EEG sensors enable non-invasive monitoring of brain activity, but the signals are noisy, high-dimensional, and uncertain. This project explores how core AI techniques—learning, probabilistic reasoning, and decision-making—can be applied to real physiological data to infer cognitive states and study how external stimuli influence them. The project highlights the challenges of building robust models from biological data. It demonstrates how these models can support systems that respond more effectively to human signals, with important applications in healthcare, education, and assistive technology. Because cognitive states change over time and differ across individuals, building AI systems that can reliably interpret EEG data requires methods that can handle both variability and uncertainty. Success in this area could enable more adaptive human-machine systems that personalize feedback, improve learning environments, and support health-related monitoring.

Students will work with EEG data collected during auditory, gustatory, and olfactory stimulation, as studied in the referenced paper, to design and evaluate AI systems that classify, predict, or reason about cognitive states.

By completing this project, students will:

- Design **intelligent agents** that reason from physiological data
- Apply **machine learning** to real-world, noisy datasets
- Use **probabilistic reasoning** to model uncertainty in brain signals
- Compare **multiple AI approaches** empirically
- Reflect on **ethical issues** involving neurodata, privacy, and AI-mediated cognitive regulation

Problem Statement: Design an AI system that **infers cognitive states** from EEG signals recorded during different sensory stimulation conditions (auditory, gustatory, olfactory).

Objectives

1. **Problem Formulation (Intelligent Agents)**
 - Define the agent's percepts (EEG features), actions (classification or prediction), and performance measure.
 - clearly specify target variables (e.g., stimulation type, cognitive state labels, engagement levels).
2. **Machine Learning**
 - Implement at least **one** decision tree and one neural network (e.g., a convolutional neural network, or a transformer architecture).
 - Extract or select meaningful EEG features (e.g., band power, statistical summaries).
3. **Probabilistic Reasoning**
 - Model uncertainty using a probabilistic approach, such as:
 - Naïve Bayes
 - Bayesian networks
 - Hidden Markov Models (temporal EEG dynamics)
4. **Comparative Analysis**
 - Compare a decision tree, a neural network, and a probabilistic reasoning model in predicting/infering the cognitive state of a person from EEG data.
5. **Ethical Reflection**
 - Discuss ethical issues: cognitive manipulation, informed consent, data privacy, bias, and misuse of neurotechnology.

Methods

The system we made was created as an intelligent agent with precepts, actions, and performance measures. The agent's percepts were EEG-derived features extracted from the recorded brain signals given in the dataset. These features include band-power measurements across specific frequency bands, such as alpha, beta, delta, etc. The agent's action was to classify the sample into the correct stimulation condition, and the performance measures were based on classification accuracy, confusion matrices, and CNN training behavior.

Two experiments were performed: in experiment 1, EEG samples were classified into music stimulation conditions, such as no music, relaxing music, exciting music, and all music combined. In experiment 2, the model also classified stimulation conditions, but whereas experiment 1 used music as the medium, experiment 2 used drinking coffee, smelling perfume, or having no stimulation. Band power histograms were created for both experiments to visually compare how the frequency band distributions changed according to the conditions.

Three main types of AI methods were implemented. First, a decision tree classifier was used as an interpretable baseline machine learning model. The decision tree learns feature-based rules that separate EEG samples into predicted classes and provides feature importance rankings. Second, a convolutional neural network to test whether a neural network model can capture more complex patterns in the EEG signal, as they are better at learning nonlinear relationships that simpler models struggle with. The CNN was trained using cross-entropy loss, and its training loss curves were recorded over 50 epochs. Third, probabilistic reasoning methods were used to model uncertainty in the EEG data. These included a Bayesian network and a Hidden Markov Model. The Bayesian network was used to model relationships and uncertainty among variables, while the HMM modeled hidden cognitive-state transitions over time.

All experiments were run locally on the same machine and with the same software environment. The required Python libraries and dependencies were listed in the requirements.txt file to ensure reproducibility of the experiments. Each model was trained and evaluated using the same processed dataset splits to ensure a fair comparison.

Experiment Design

Our experiments used EEG data from the dataset linked at the bottom of the page, with the main data comprising EEG-derived features across multiple channels and frequency bands. We extracted these features from the epochs and organized them into structured samples, where each sample corresponded to a specific condition, and the labels depended on the current experiment.

In experiment 1, the target variable was the music stimulation condition. The classes included categories, such as no music, relaxing music, exciting music, and combined music. In experiment 2, the target variable was the non-music stimulation condition. The classes were tasting coffee, smelling perfume, and providing no stimulation.

For both experiments, the EEG band power distributions were plotted for channel AF7 across the delta, theta, alpha, beta, and gamma bands. These plots were used to assess whether the stimulation conditions produced visually separable EEG patterns.

The main models we used during evaluation were the Decision Tree Classifier, the Convolution Neural Network, the Bayesian Network, and the Hidden Markov Model. The decision tree served as the baseline model for interpretation, as it classified EEG samples using feature-based decision rules. We also created feature importance plots to identify which variables had the strongest influence on the tree's predictions, and we used the CNN to capture more complex nonlinear patterns in the data. We used 50 epochs for training, and recorded the cross-entropy loss to assess how well the model learned over time. The

Bayesian network was used to represent uncertainty and relationships among the variables and labels. The HMM was used to model transitions in hidden cognitive states over time. The dataset was split into 80/20 training/test sets.

During the experiment, the decision tree and CNN were evaluated using 5-fold cross-validation accuracy, macro AUC-ROC, and several normalized confusion matrices. Accuracy was measured as the overall percentage of correct predictions, and Macro AUC-ROC measured how well each model separated the classes across thresholds, treating each class equally. The confusion matrices were used to examine which classes were classified correctly and which classes were most commonly confused.

Results and Discussion

The results show that EEG-based classification is possible but difficult due to the noisy, overlapping nature of EEG signals. The band power histograms for both experiments show that the stimulation conditions were not clearly separated across the alpha, beta, delta, theta, and gamma bands, and many distributions overlapped, indicating that the models had to learn more subtle differences rather than obvious differences between classes.

In Experiment 1, the CNN achieved the highest classification accuracy of 0.672, while the decision tree achieved 0.606. However, the decision tree achieved a higher macro AUC-ROC score of 0.872 than the CNN's, 0.841. This means that the CNN made more accurate final predictions overall, but the decision tree exhibited slightly stronger class separation.

The confusion matrices for Experiment 1 show that both models had strengths and weaknesses. The decision tree performed well on some classes, such as the exciting music and no-music categories, but it confused several all-music samples with exciting music. The CNN performed strongly on the all-music and no-music classes, but it struggled more with relaxing music and exciting music. This suggests that some music-related EEG conditions may produce similar signal patterns, especially when the brain responses to different types of auditory stimulation overlap.

The CNN training loss for Experiment 1 decreased steadily from approximately 1.18 to approximately 0.83 over 50 epochs, indicating that the CNN was accurately learning patterns from the training data. However, because the final accuracy was still moderate, the model may have been limited by class noise or feature limitations.

In Experiment 2, the CNN performed much better than the decision tree. The CNN achieved an accuracy of 0.837 and a macro AUC-ROC score of 0.888, whereas the decision tree achieved an accuracy of 0.468 and an AUC-ROC score of 0.737. This shows that the CNN was much more effective for the classification task in experiment 2 than in experiment 1.

The Experiment 2 confusion matrices show that the CNN classified coffee samples very well, but it still struggled with perfume. Many perfume samples were misclassified as either coffee or no stimulation, suggesting that the EEG response to perfume may have been less distinct than that to coffee, or that the selected features did not fully capture the difference between the two stimulations. The decision tree also struggled in this experiment, especially because it tended to classify many samples as no stimulation. These points show that the decision tree may not have been sufficiently flexible or complex to model the more complex patterns in the actuation dataset. The CNN training loss for Experiment 2 decreased from approximately 0.75 to approximately 0.43 over 50 epochs. This strong decrease, along with the higher accuracy and AUC-ROC, suggests that the CNN model learned more useful patterns in Experiment 2 than in Experiment 1, and the actuation classes may have been easier for the CNN to separate, or the features may have more effectively represented those conditions.

In both experiments, the decision tree feature importance plots showed that `epoch_start_ts` was the most important feature. It was the most dominant feature, with almost all the other features contributing very little. This suggests that the decision tree may have relied heavily on epoch-order information rather than on EEG patterns. This can reduce generalization to new variables or differently ordered data.

The HMM showed how the model assigned hidden states across the timeline, while the probability plot showed variance and uncertainty in the state assignments. The transition matrix indicates that some hidden states were stable. One example is that state 0 has a high probability of 0.906, meaning the model stayed in state 0 once it entered that state. This is also shown in the other 3 states. Overall, the CNN was the strongest classification model.

Conclusion

This project explored how artificial intelligence can infer cognitive states from EEG signals and how it treated the problem as an intelligent-agent task, in which EEG features served as percepts, where the agent's action was to classify the stimulation condition. The project compared several AI approaches, including a decision tree, a convolutional neural network, a Bayesian network, and a Hidden Markov Model.

The results showed that EEG-based classification is possible but challenging. The band power histograms show that the distributions across conditions overlap heavily, and that the EEG data is noisy across conditions and participants. In experiment 1, the CNN achieved the highest accuracy of 0.672. In contrast, the decision tree achieved the highest AUC-ROC of 0.872, suggesting that the CNN made the best overall predictions, but the decision tree best separated the classes. In experiment 2, the CNN was also the best model, recording an accuracy of 0.836 and an AUC-ROC of 0.888. The CNN's training loss also decreased

steadily, showing that it was learning meaningful patterns from the data. The confusion matrix showed that some classes were still difficult to classify correctly.

The probabilistic reasoning models added important uncertainty-based data to the project. The Bayesian network modeled uncertainty in features and labels, while the HMM modeled hidden-state transitions over time. The HMM's transition matrix showed that some hidden states were relatively stable, meaning the model often remained in the same state over time.

A major limitation of the project is that the `epoch_start_ts` feature dominated the feature-importance plots in the decision tree. This can mean that the decision tree relied heavily on ordered information from the experiment. This can reduce the model's ability to generalize to new participants or differently ordered experiments. Future work should repeat the experiments with time-based features removed or controlled to test whether the models can classify conditions using only EEG-based physiological features.

Future work could improve the system by employing stronger EEG preprocessing, more effective artifact removal, and more advanced feature extraction. Additional models, such as other neural networks and transformer-based models, could be tested to capture EEG patterns better. The system could also combine EEG with other physiological signals from the dataset, such as heart rate or reaction time. This multimodal approach may improve classification because cognitive state is not always fully observable from the EEG data alone.

We noted that this project exists in a grey area of AI ethics, as EEG data are sensitive neurodata and can reveal information about a person's attention, stress, fatigue, or emotional state. Any AI system designed to use EEG data must take steps to protect user privacy, obtain informed consent, and avoid using these predictions to manipulate or monitor any person. While AI-based cognitive monitoring has extremely promising applications in healthcare, assistive technology, and human-computer interaction, it must be used responsibly to ensure that any users remain informed, protected, and in control of their data.

References:

[1] H. Fekri Azgomi, Luciano R. F. Branco, Md Rafiul Amin, Saman Khazaei, and Rose T. Faghih. "Regulation of brain cognitive states through auditory, gustatory, and olfactory stimulation with wearable monitoring." *Scientific reports*, vol. 13, no. 1, p. 12399, 2023.

Dataset: <https://physionet.org/content/brain-wearable-monitoring/1.0.0/>

GitHub: https://git-classes.mst.edu/jsszdd/cs5400_semester_project

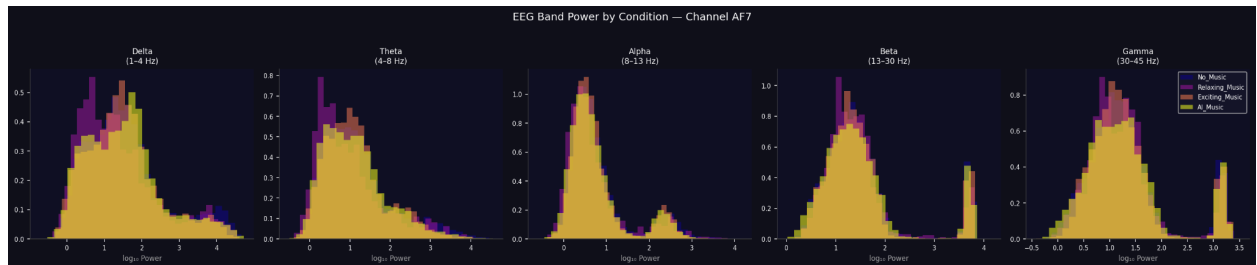
Contributions:

Signed off by: Nathanael, James, Sai

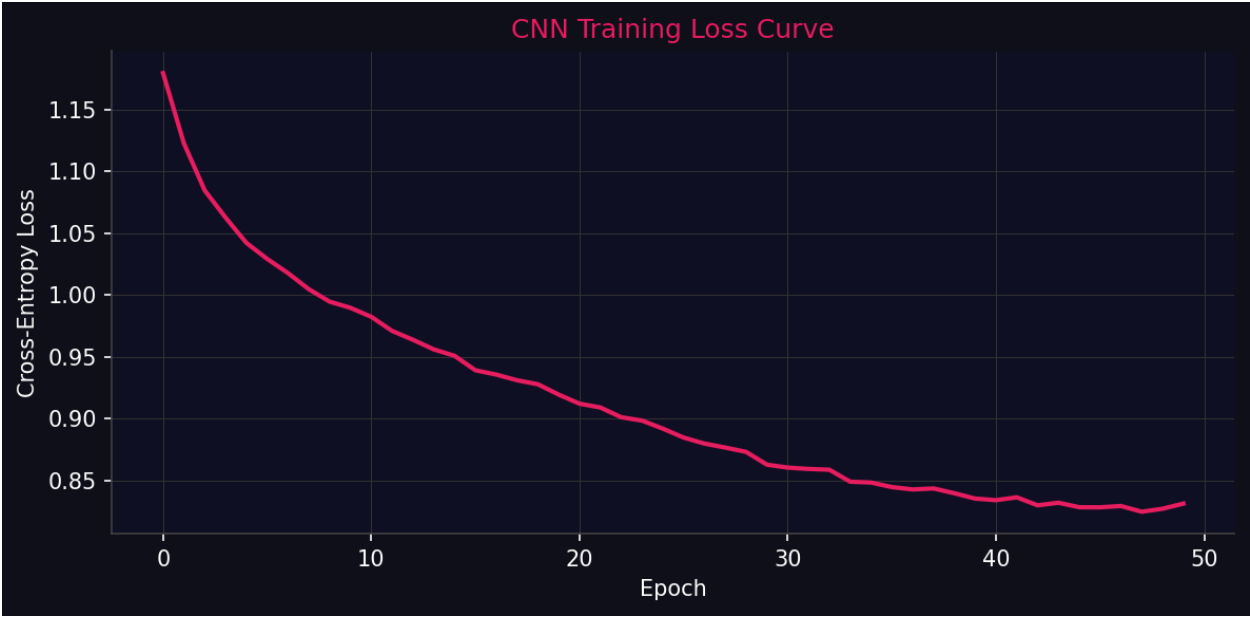
Each member of the team did roughly the same amount of work, although Nathanael joined the team late, we communicated effectively to assign tasks for each of us to complete. While Nathanael and I worked mainly on the code, Sai still helped quite a bit and picked up the slack in writing the report. Sai created the template and wrote the outline of the report, Nathanael and I designed a decent chunk of the coding portion, and Sai finished up the main processes after the skeleton was done. Then Nathanael and I did the Quad chart with Sai writing out the rest of the experiment design and the methods pages. We all believe we did equal work and worked well as a team.

Experiment 1 references

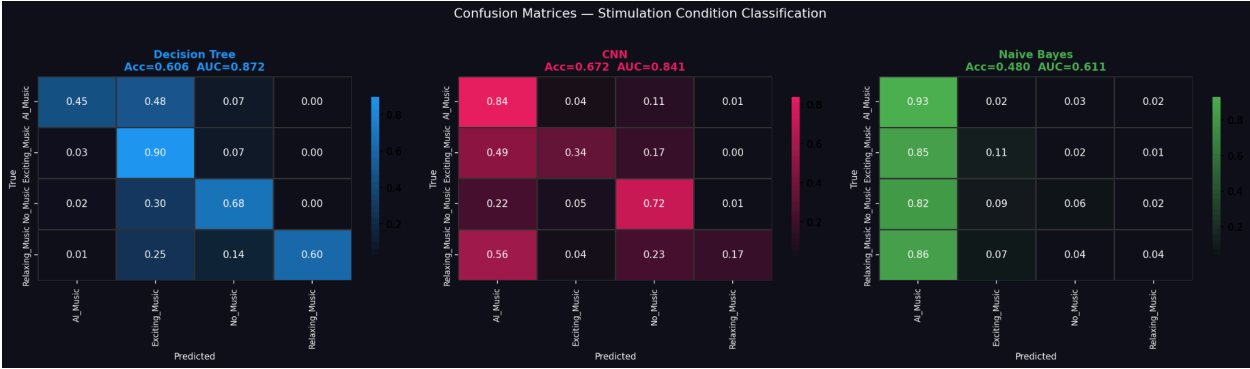
Appendix A



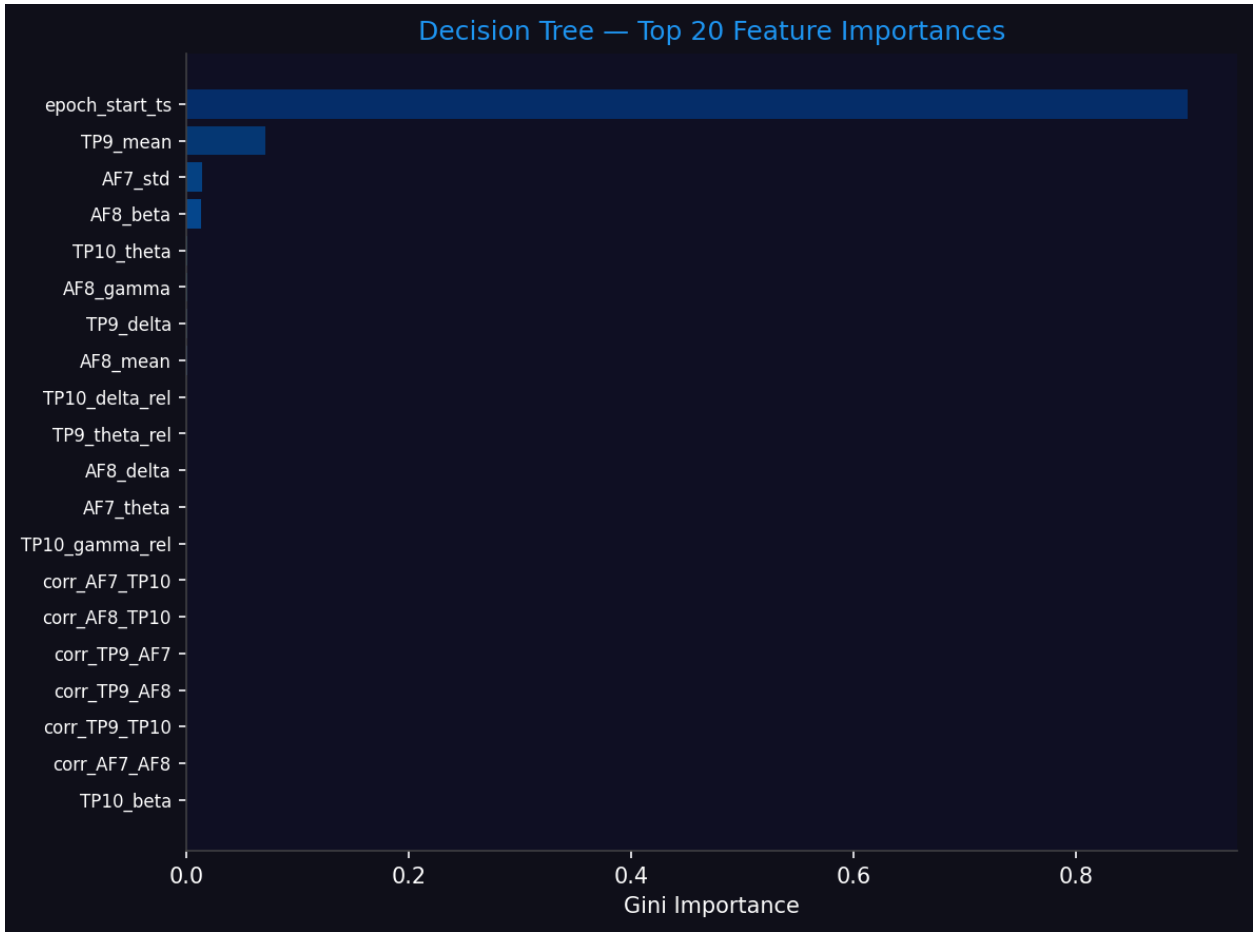
Appendix B



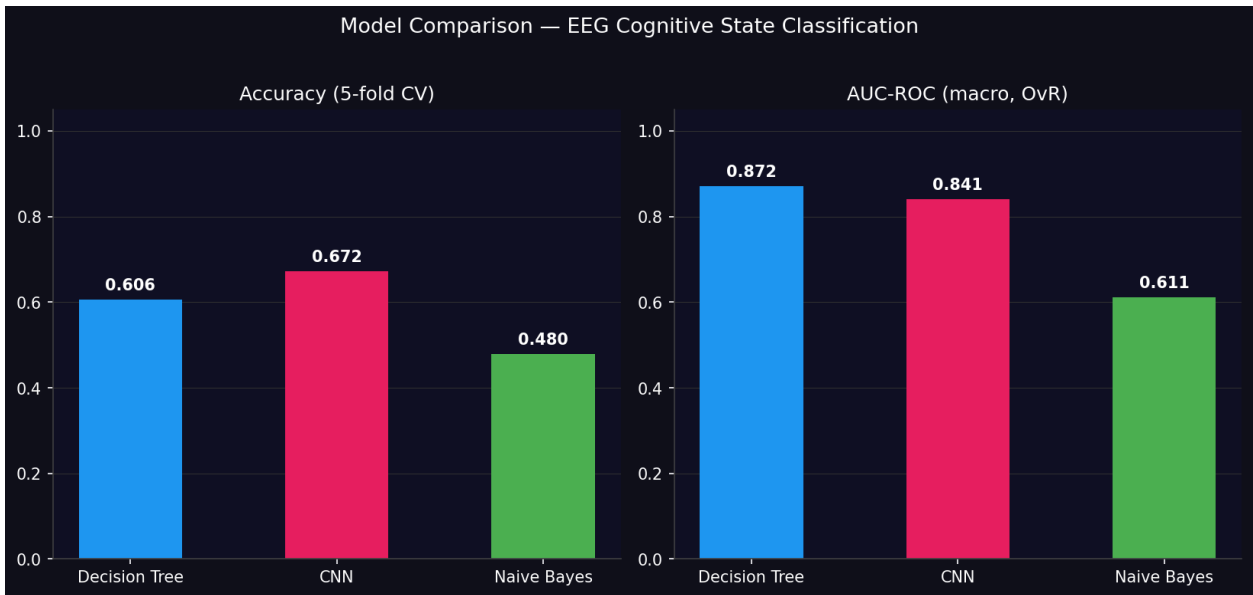
Appendix C



Appendix D

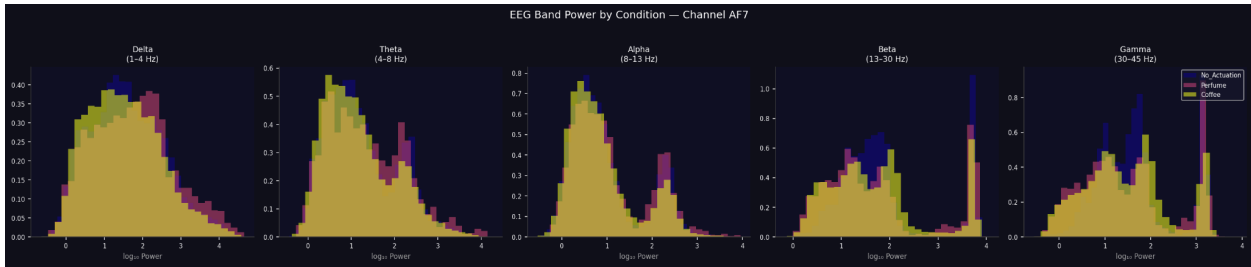


Appendix E



Experiment 2 references

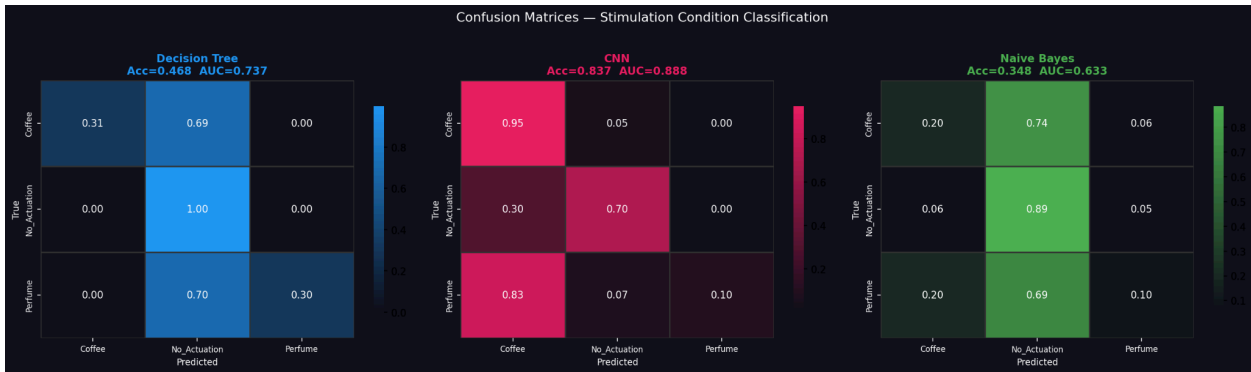
Appendix F



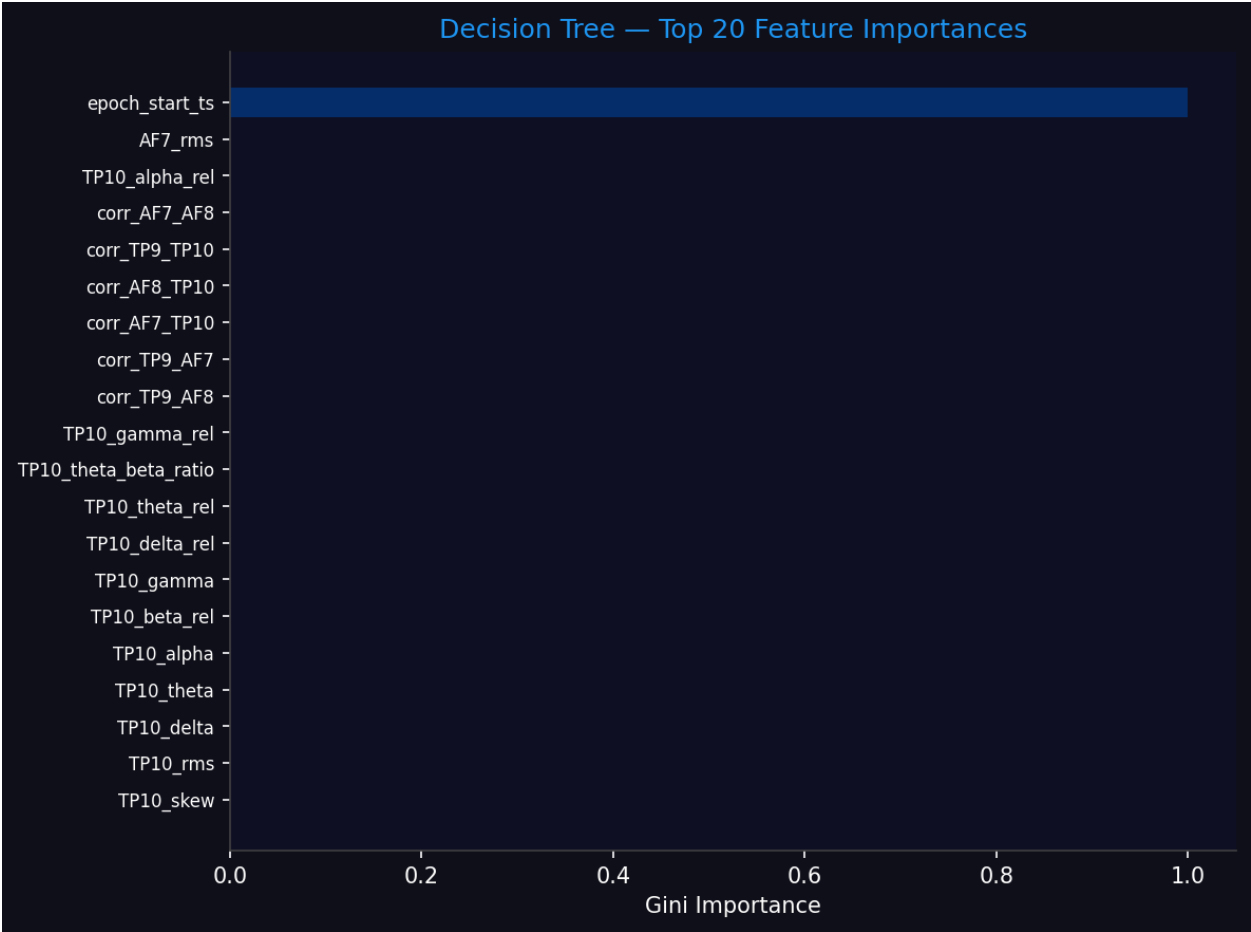
Appendix G



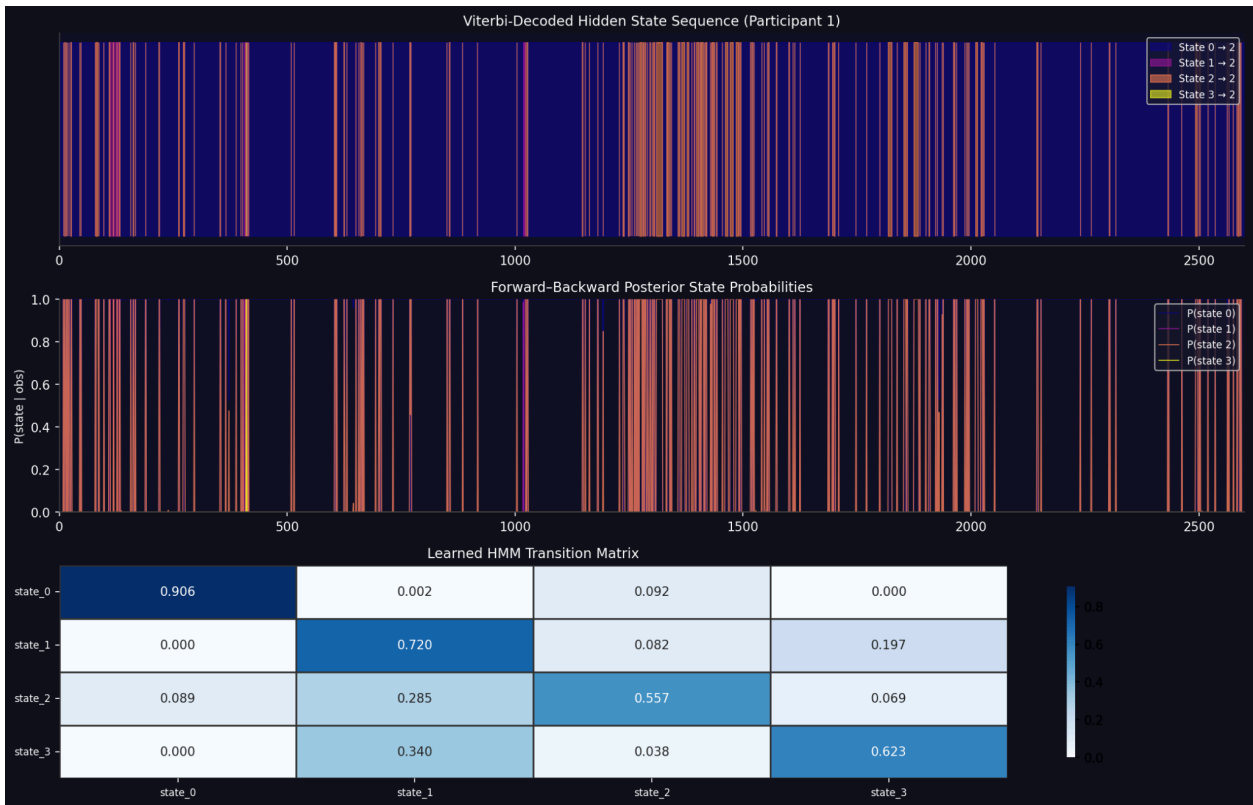
Appendix H



Appendix I



Appendix J



Appendix k

